KENYA MEDICAL TRAINING COLLEGE
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF CLINICAL MEDICINE

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN CLINICAL MEDICINE & SURGERY

PAPER: SURGERY

DATE: 4th July 2019	**TIME: 3 HOURS (9:00AM – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION A: Multiple Choice Questions (MCQs) - 40 MARKS

1. Avascular necrosis:
   a) Means death of blood vessels
   b) Is commonly encountered in skull fractures
   c) Is death of bone from deficient blood supply
   d) Is caused by osteoporosis

2. A military recruit who began training six weeks ago comes to the hospital with complaints of pain in his right mid foot for one week. The pain is increasing in intensity day by day. There is no history of obvious trauma to the foot. X-ray reveals a fracture in his third metatarsal. He most likely has
   a) A pathological fracture
   b) A fatigue fracture
   c) A segmental fracture
   d) A traumatic fracture

3. Which one of the following is characteristic of osteogenic sarcoma?
   a) Onion peel appearance
   b) Multiple punched out lesions
   c) The tumour is capped with cartilage
   d) Sun-rays appearance

4. In acute haematogenous osteomyelitis, the most likely causative organism is
   a) Beta haemolytic streptococcus
   b) Salmonella typhi
   c) Staphylococcus aureus
   d) Neisseria gonorrhea

5. Regarding bulbocavernosus reflex in a spinal injury patient with acute paralysis:
   a) Presence of the reflex indicates spinal cord severance
   b) Presence of the reflex indicates presence of head injury
   c) Absence of the reflex indicates obvious complete cord transection
   d) Absence of the reflex indicates that there is no spinal injury

6. Which one of the following statements is not true regarding surgical jaundice?
   a) Carcinoma of the head of pancreas is a cause
   b) Urine test for urobilirogen is usually positive
   c) Patients usually present with itchiness of the skin

d) Patients pass pale stool

7. Which one of the following statements correctly presents brain concussion in head injury?
   a) It is the commonest cause of persistent coma in head injury
   b) It causes tears of the neuron in the brain
   c) It is a bruise of the brain
   d) It is a mild brain injury that temporarily impairs brain functions

8. The following is true regarding hydrocephalus
   a) It presents with sunken fontanelles
   b) Overriding of skull bones is diagnostic
   c) Endoscopic third ventriculostomy is an effective method of treatment
   d) Stunting is mostly done on the left side of the head

9. A surgery that is performed to fuse a joint is termed
   a) Arthrodesis
   b) Arthrotomy
   c) Arthroscopy
   d) Arthroplasty

10. A patient with history of passage of hard faeces associated with severe burning, tearing, sharp pain in the anal region, most likely has;
   a) Hirschsprungs disease
   b) Fissure - in - ano
   c) Fistula – in - ano
   d) First degree haemorrhoids

11. The most common causative organism in cellulitis is;
   a) Pseudomonas aeruginosa
   b) Streptococcus pneumonia
   c) Staphylococcus epidermidis
   d) Beta haemolytic streptococcus

12. Right sided pleural effusion presents with;
   a) Tracheal deviation towards the right
   b) Hyper resonant note on percussion of the right chest
   c) Stony dullness on the right chest
   d) Good air entry bilaterally

13. Tension band wiring is used in;

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a) displaced patella fractures
b) Fracture of the neck of femur
c) Fractured ribs
d) Fracture of the scaphoid bone

14. An epiphyseal injury in which there is separation of the epiphysis with a characteristics triangular fragment of the metaphysis attached to the displaced epiphysis is classified as salter- Harris type;
    a) I
    b) Ii
    c) Iii
    d) Iv

15. An antimicrobial that is used in surgical patients to cater for infection by anaerobic organisms is;
    a) Ceftriaxone
    b) Gentamycin
    c) Flucloxacillin
    d) Metronidazole

16. Which one of the following statements does not correctly apply in congenital hypertrophic pyloric stenosis?
    a) It is commoner in first born male infants
    b) The patient has projectile vomiting
    c) The vomitus is bilious and occurs shortly after feeding
    d) Presents within the first eight weeks of life

17. A pre-operative medication that is used to reduce secretions to prevent aspiration during general anesthesia is
    a) Pethidine
    b) Atropine
    c) Diazepam
    d) Omeprazole

18. One of the following indications for amputation is classified as constituting a deadly extremity
    a) Severe contracture
    b) Dry gangrene from peripheral vascular disease
    c) Malignant tumour in a limb
    d) Congenital abnormality such as polydactyl

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19. A fracture of the upper/proximal end of ulna with dislocation of the head of radius is called
    a) Galeazzi fracture-dislocation
    b) Smith's fracture
    c) Cole's fracture
    d) Monteggia fracture-dislocation

20. Free gas underneath the diaphragm seen in a plain abdominal X-ray is indicative of
    a) Pneumothorax
    b) Perforated abdominal viscus
    c) Pleural effusion
    d) Ascites

21. A congenital abnormality in which the urethral meatus is located on the dorsal aspect of the penile shaft is termed as
    a) Epispadias
    b) Phimosis
    c) Paraphimosis
    d) Hypospadias

22. A hyper resonant percussion note would be produced by which of the following conditions
    a) Empyema thoracis
    b) Haemothorax
    c) Pneumothorax
    d) Pleural effusion

23. A patient who sustains injury of the right lower limb after a road traffic accident presents with pain in the hip area and inability to use the right lower limb. On examination, the limb is shortened, adducted and internally rotated. The patient most likely has
    a) Fracture neck of femur
    b) Posterior dislocation of the hip
    c) Trochanteric fracture of femur
    d) Central fracture –dislocation of the hip

24. Which one of the following statements is true regarding management of fresh cut wounds?
    a) Ketamine is used as a local anaesthetic
    b) Suture size 1/0 is best for the face
    c) Hydrogen peroxide is recommended for cleaning of fresh wounds
    d) Lignocaine with adrenaline should not be used to anaesthetize fingers

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25. Regarding swallowed foreign bodies, which one of the following statements is not true
   a) Urgent intervention is required if the ingested object is sharp
   b) A swallowed high powered magnet is inconsequential
   c) A coin in the lower oesophagus may be observed for 24 hours to monitor its passage into the stomach
   d) Foreign body in the oesophagus can be removed endoscopically

26. About the deformities in congenital Talipes Equinovourus
   a) The foot is excessively dorsiflexed
   b) The foot is in eversion
   c) The Achilles tendon is usually shorter than the normal side
   d) The leg is usually in lateral rotation

27. The following are complications of fracture shafts of radius and ulna, except
   a) Avascular necrosis
   b) Cross union
   c) Non- union
   d) Compartment syndrome

28. The surgical term used to denote excision of an organ ends in –
   a) Otomy
   b) Ostomy
   c) Ectomy
   d) Oplasty

29. The following are complications of duodenal ulcer, except
   a) Perforation
   b) Malignant change
   c) Haemorrhage
   d) Gastric outlet obstruction

30. Which one of these hernias frequently strangulates?
   a) Direct inguinal hernia
   b) Umbilical hernia
   c) Hiatus hernia
   d) Indirect inguinal hernia

31. Osteosclerotic deposits in lumbar spine could be due to:
   a) Metastasis from breast cancer
   b) Metastasis from prostatic cancer

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c) Metastasis from renal malignancies
d) Metastasis from cancer of the stomach.

32. An investigation used for staging carcinoma of the oesophagus is:
a) Oesophagoscopy
b) Endoscopic ultrasound
c) Barium swallow
d) Barium meal

33. Regarding spina bifida
a) Spina bifida occulta is obvious on the skin surface
b) Rachischisis has nerve elements openly exposed
c) The condition has no relationship with lower limb deformities
d) All cases present with neurological deficits

34. Pyomyoscitis
a) Presents with blistering
b) Is inflammation of tendons
c) Is accumulation of pus in muscles
d) Is treated with antibiotics only.

35. Compartment syndrome following fracture is treated surgically by
a) Escharotomy
b) Incision and drainage
c) Arthrotomy
d) Fasciotomy

36. Which one of the following fluids is a colloid?
a) Haemacel
b) Normal saline
c) Ringer's lactate
d) 5% dextrose

37. Regarding carcinoma of the breast
a) Peau d'orange is a description of the consistency of the tumour.
b) Breast cancer is found most frequently in the upper outer quadrant
c) Most breast cancers present as cystic swellings
d) Chest radiography are not necessary

38. Complications of gallstones include all these, except

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a) Acute cholangitis
b) Intestinal obstruction
c) Acute appendicitis
d) Acute pancreatitis

39. Acute intussusception
a) Is a twisting of a loop of intestine
b) Is most common in adults
c) Occurs when a portion of intestine is invaginated within an adjacent segment
d) Is not a known cause of intestinal obstruction

40. The Ochsner-Sherren regime is used in
a) Management of an appendicular mass
b) Management of full-thickness rectal prolapse
c) Management of chemical burns
d) Management of carcinoma of the urinary bladder

SECTION 2: SHORT ANSWER QUESTIONS (SAQs) – (40 MARKS)

1. State five (5) causes of non- union of a fracture **(5 marks)**
2. Describe five (5) pathological changes that occur in a persistently dislocated hip joint **(5marks)**
3. A patient is diagnosed with 3rd degree haemorrhoids. State five (5) possible complications of haemorrhoids this patient could develop. **(5 marks)**
4. Outline five (5) aetiological factors that contribute to the causation of abdominal hernias **(5 marks)**
5. State five (5) features of chronic osteomyelitis as seen on a radiograph of a long bone **(5 marks)**
6. An adult female patient presents to you with a neck swelling that you believe is the thyroid gland. You notice features suggestive of hyperthyroidism. State five (5) of these features. **(5 marks)**

7. A 30-year-old male weighing 50kg is brought to causality with burns of the trunk and thighs anteriorly. you estimate the percentage to be 25%. Clearly outline your fluid resuscitation of this patient in the first 24 hours using parkland's formula. **(5 marks)**

8. An adult patient is brought to hospital with paraplegia. State five (5) condition or injuries of the vertebral Colum that could cause compression or injury to the spinal cord or cauda equina. **(5 marks)**

SECTION 3: LONG ANSWER QUESTION (LAQs) – (20 MARKS)

1. A motorcycle rider is knocked down by a speeding vehicle on a busy highway. He sustains multiple injuries including an open severely comminuted fracture of the tibia and fibula of his right leg. Describe his management from the time he presents to causality in a county referral hospital to the time of discharge. **(20 marks)**

Primary survey.

Airway

Breathing

Circulation

Arrest bleeding

Resuscitate the

Secondary survey

T.T.

Antibiotics

Investigations

Fix a. I.V line

Analgesics

Admit